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THE EPIDEMIOLOGY OF RHEUMATOID ARTHRITIS

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Epidemiology is the study of the distribution and determinants of disease in human populations.²¹ This definition is based on two fundamental assumptions: first, that human disease does not occur at random and, second, that human disease has causal and preventive factors that can be identified through systematic investigation of different populations or subgroups of individuals within a population in different places or at different times. As a result, epidemiologic studies include simple descriptions of the manner in which disease appears in a population (i.e., levels of disease frequency as well as incidence and prevalence, mortality, trends over time, geographic distributions, and clinical characteristics) and descriptions of the role of putative risk factors for disease occurrence. Incidence studies include all new cases of a specified condition arising in a defined population over a specified time period, and prevalence studies include all patients with the condition who are present in a population at a particular point in time. As shown in [Figure 1](#), prevalence cohorts exclude patients who died or left the population soon after their incidence date, and they include patients arising in different populations who moved into the cohort after their incidence date. Because of this, there is a greater potential for bias to be

introduced in prevalence cohorts as compared to incidence cohorts. Thus, population-based incidence cohorts are superior to prevalence cohorts for descriptive epidemiologic studies.

Epidemiologic studies of risk factors fall into three major categories: prospective cohort studies, retrospective cohort studies, and case-control studies. The relation between these is illustrated in [Figure 2](#). In a prospective cohort study, a study population is assembled, none of whom have experienced the outcome of interest, and is followed forward into the future. People in the cohort are classified according to those characteristics that might be related to outcome, that is, putative risk factors. These people are then observed over time to determine which of them experience the outcome. The analysis addresses the question of whether people who were exposed to the risk factor were more likely to develop the outcome compared with those who were not exposed. In a retrospective cohort study, the cohort of individuals is identified from past records and followed forward up to the present. Data regarding historical exposure to the putative risk factor are collected retrospectively, typically by examination of medical records. As in a prospective cohort study, retrospective cohort studies also compare the frequency of the outcome in exposed patients as compared to unexposed patients. In a case-control study, two cohorts are assembled, one that has the outcome of interest and another that is free of the outcome of interest. Data regarding exposure to the putative risk factor in the two groups are collected retrospectively so as to determine whether patients with the outcome of interest were more likely to have had a history of the exposure of interest compared with those who were free of the outcome of interest. Of these three study designs, prospective cohort studies have fewer potential biases than the other two; however, they are frequently not feasible because they typically require extended follow-up, often 5 to 10 years or more into the future. A detailed comparison of the potential biases involved in retrospective cohort studies and case-control studies is beyond the scope of this article but can be readily found in numerous epidemiologic publications.[21](#), [30](#), [61](#) In this article, we review data on the descriptive epidemiology (incidence, prevalence, and survival) and risk factors associated with rheumatoid arthritis (RA).

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Section snippets

Incidence

The most reliable estimates of incidence, prevalence, and mortality in RA are those derived from population-based studies. Several of these have been conducted in a variety of geographically and ethnically diverse populations. The Norfolk Arthritis Register is a prospective population-based database that was established to study new cases of inflammatory arthritis as they occurred in the community and to follow them prospectively to investigate the natural history of the condition. This data ...

RISK FACTORS ASSOCIATED WITH RHEUMATOID ARTHRITIS

A number of risk factors have been suggested as important contributors to the development or progression of RA. Of these, the best studied have been genetics, infectious agents, oral contraceptive medications, smoking, and formal education. ...

SUMMARY

Studies of the descriptive epidemiology of RA indicate a population prevalence of 0.5% to 1% and a highly variable annual incidence (12–1200 per 100,000 population) depending on gender, race/ethnicity, and calendar year. Secular trends in RA incidence over time have been shown in several studies, supporting the hypothesis of a host-environment interaction. People with RA have a significantly increased risk of death compared with age- and sex-matched controls without RA from the same community. ...

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a risk factor associated with RA. ...

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